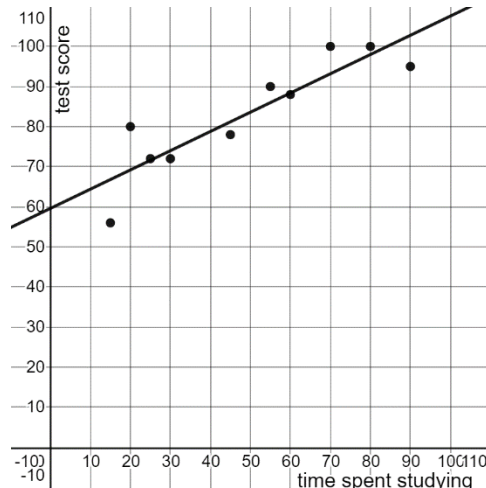


Algebra 1
Unit 3
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

For questions 1 through 5, the scatter plot shows the relationship between the time students spent studying, in minutes, for a test and the test score earned.



The equation $y = 0.48x + 59.58$ can be used to model the relationship between the time spent studying in minutes, x , and the test score earned, y .

1. Find the slope of the model.

0.48

2. Determine the y -intercept of the model.

59.58

3. What does the slope represent in the context?

The test score will increase by 0.48 points per minute of study.

4. Determine an appropriate domain in the context.

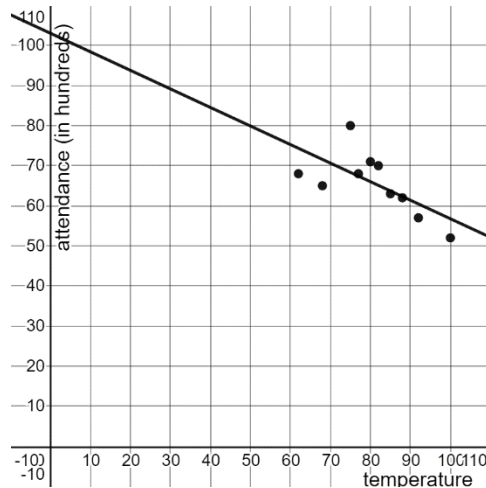
$\{x | 0 \leq x \leq 84.2\}$

$$\begin{aligned} 100 &= 0.48x + 59.58 \\ -59.58 &\quad -59.58 \\ \hline 40.42 &= \frac{0.48x}{0.48} \\ 84.2 &= x \end{aligned}$$

5. Determine an appropriate range in the context.

$\{y | 59.58 \leq y \leq 100\}$

For questions 6 through 10, the scatter plot shows the relationship between the average daily temperature and the daily attendance at a theme park.



The equation $y = -0.46x + 103.04$ can be used to model the relationship between the temperature, x , and the attendance at a theme park, y .

6. Find the slope of the model. Show your work.

-0.46

7. Determine the y -intercept of the model.

103.04

8. What does the slope represent in the context?

Attendance will drop by 0.46 hundred people per degree of temperature increase.

9. Determine an appropriate domain in the context.

Answers may vary.

Sample answer: $\{x | 50 \leq x \leq 100\}$

10. Determine an appropriate range in the context.

$\{y | y \geq 0\}$